# Ssu-Yuan Lin (Blake)

**AI Software Engineer**

ssu.yuan.lin@gmail.com · github.com/BlakeLin-LSY · blakelin-lsy.github.io/web-resume

Software and algorithm engineer with 3+ years of production experience across computer vision, hardware-software integration, and high-performance ML workflows. Track record spans AI agent infrastructure, real-time 3D optical metrology, and factory automation.

## Professional experience

**Independent AI & Agent Infrastructure Engineering** 2026-05 – Present

Architected Agent Ticket Kanban (ATK) state-machine task dispatcher (1,080+ tests) and windowphase context-injection CLI (69.5% input token reduction).

**DynaSafe Corp.** · Senior AI Engineer 2026-01 – 2026-04

- Profiled PhysicsNeMo dataloader I/O bottlenecks and directed Claude-agent pipeline refactoring, cutting training cycle time by ~10×.
- Engineered real PCAP network-anomaly detection workflows using Nvidia Morpheus; adapted PhysicsNeMo for OpenFOAM CFD simulations.

**MetGauge Co., Ltd.** · Software Engineer 2023-09 – 2024-08

- Architected C++/Python pipeline and hardware trigger sync for a 24-camera (12 stereo pairs) system, advancing MVP to factory testing.
- Formulated thermal-drift calibration specs and boundary test suites across continuous production runs, de-risking factory integration.

**Cloud Architecture & Distributed Systems · Google Grow with Google Program** 2022-04 – 2023-08

GCP infrastructure, Kubernetes (GKE) container orchestration, microservices, and continuous deep-learning study group.

**UTECHZONE Corp.** · Algorithm Engineer 2021-06 – 2022-03

- Developed C# automated optical inspection (AOI) algorithms that reduced manufacturing cycle time by 40%.
- Mentored two engineers and collaborated with a cross-functional team of 10+ on integration milestones.

**Computer Science Foundations & Self-Directed Learning** 2020-04 – 2021-05

CS50 online course; trail running, marathon endurance training, and co-founding weekly deep learning paper study group.

**LIPS Corporation** · Software Engineer, Algorithm Development 2019-07 – 2020-04

- Improved real-time 3D measurement accuracy by 66% through algorithm redesign and Intel RealSense D415/D435 calibration.

## Personal projects

**Study Atlas**

Engineered an automated research reader compiling papers into bilingual (EN/ZH) knowledge cards with dual reading tracks; implemented validation pipelines to reject over-budget plans prior to delivery.

**Transcribe-for-X**

Engineered an on-device Android speech prototype with cryptographic audio verification before local ASR dispatch, keeping raw immutable audio separate from non-destructive transcript cleanup.

## Technical skills

Python · C++ · C# · ML pipelines · Computer vision

Profiling · Stereo camera calibration · Hardware integration · Test design

## Education & earlier research

**M.S. in Physics** · National Central University · 2016 – 2018

**B.S. in Physics** · National Central University · 2012 – 2016

Research Assistant · National Central University · 2018-09 – 2019-01